

Which token statistics make machine text detectable according to GLTR?

1906.04043

Abstract

Introduction

Intro

Introduction

As generative language models grow increasingly fluent, the question of whether a machine—not a human—wrote a given passage has become both a critical forensic problem and an unsolved open challenge: text-detection benchmarks and deployed classifiers are often brittle and opaque to human reviewers. This survey examines the thesis that machine-generated text can be *reliably* detected and *made human-inspectable* through purely statistical cues (arXiv:1906.04043): because generative models sample tokens from their own predicted distributions, generated text concentrates probability mass on high-ranked tokens—a signature that supports both automated discrimination and human review. Empirically, unaided annotators detected generated text at only 54.2% accuracy, near chance, which rose to 72.3% when they used the GLTR visualization interface, while a logistic-regression discriminator confirmed that a word falling outside the model’s top-100 predictions is over 5 times more likely in human text (arXiv:1906.04043).

We first introduce the statistical rationale and the rank-based detection features underlying GLTR; next review its automated empirical validation and the human-subjects interface study; then discuss limitations, adoption impact, and implications for future detectors.

中文速览：生成模型按自身预测分布逐 token 采样，使生成文本的概率质量集中于高排名词元，这一统计签名既可供机器自动判别、亦可经可视化供人工复核。无工具时人工识别准确率仅 54.2%（接近随机），使用 GLTR 界面后升至 72.3%；逻辑回归判别器亦证实，落在模型 top-100 预测之外的词在人类文本中的出现概率高出 5 倍以上 (arXiv:1906.04043)。下文依次介绍统计原理与基于排名的检测特征、自动化验证与人机界面实验、局限性与影响，以及对后续检测器的启示。

Problem and Motivation

The detection of text generated by modern neural language models demands urgent attention because such output has become difficult for humans to distinguish from human-written prose (arXiv:1906.04043). Gehrmann et al. frame the problem not only as an automatic classification task but also as a visual analysis task, targeting end users rather than researchers alone (arXiv:1906.04043). Their motivation rests on a measurable failure of human judgment: in a human-subjects study, participants without the GLTR interface detected fake text with only 54.2% accuracy—barely above random chance—which rose to 72.3% when the interface was used (arXiv:1906.04043).

Empirically, the difficulty and its remedy both trace to how language models sample. A logistic-regression discriminator built on GLTR rank features found that a word outside the top-100 predictions is over five times more likely in real text (odds ratio 5.32), while being the top prediction is far rarer (odds ratio 0.09) (arXiv:1906.04043). In other words, generated text concentrates probability mass on high-rank tokens because models sample from their own predicted distributions, whereas human-written text uses high-rank, non-obvious words regardless of the estimated entropy (arXiv:1906.04043). This statistical signature is what GLTR exposes to the eye. The acute need for accessible detection is underscored by the community response: within its first month of public deployment, GLTR’s demo received 30,000 page views and its blog 21,000 (arXiv:1906.04043).

The open gap that remains is that these results come from a single detector evaluated on its own generation of language models, so it is unclear whether the statistical cues and human-interface gains generalize to larger, instruction-tuned models or to adversarial attacks—a question later sections take up.

问题与动机

现代神经语言模型生成的文本与人类书写文本日益难以区分，检测这类机器生成文本因此成为当务之急 (arXiv:1906.04043)。Gehrmann 等人将这一任务不仅框定为自动分类问题，也视为视觉分析任务，把目标用户定位为最终用户而非仅仅是研究者 (arXiv:1906.04043)。这一动机建立在人类判断失误的量化证据上：在一项被试研究中，未使用 GLTR 界面的参与者识别伪造文本的准确率仅为 54.2%——仅略高于随机水平——而在使用 GLTR 后升至 72.3% (arXiv:1906.04043)。

实证上，困难与其解法都源于语言模型的采样方式。基于 GLTR 排序特征构建的逻辑回归鉴别器发现，一个词落在前 100 名预测之外时出现在真实文本中的几率 (odds ratio) 高达 5.32，而成为最高预测 (top-1) 的几率则低至 0.09 (arXiv:1906.04043)。换言之，生成文本将概率质量集中在高排序词元上，因为模型从自身预测分布中采样；而人类文本无论估计熵高低，都更频繁地使用高排序、非显而易见的词 (arXiv:1906.04043)。这正是 GLTR 以视觉方式呈现的统计特征。社区对这一需求的高度关切不难佐证：公开部署后的第一个月内，GLTR 的演示页面获得 30,000 次浏览量，博客获得 21,000 次 (arXiv:1906.04043)。

由此留下的开放缺口是：上述结论来自单一检测器，且是在当时一代语言模型的语境下获得，统计线索与人类界面的增益能否推广到更大规模、经指令微调的模型或对抗性攻击，仍属未知——这正是后续章节要处理的问题。

Authorship and Provenance of the Work

GLTR originates as a 2019 preprint, arXiv:1906.04043, authored by Sebastian Gehrmann (Harvard SEAS), Hendrik Strobelt (IBM Research / MIT-IBM Watson AI Lab), and Alexander M. Rush (Harvard SEAS) (arXiv:1906.04043). Its central thesis is that machine-generated text can be reliably detected and made human-inspectable through statistical cues: because generative language models sample tokens from their own predicted distributions, generated text concentrates probability mass on high-rank tokens, which a statistical detector plus visual coloring can expose (arXiv:1906.04043).

Empirically, a logistic-regression discriminator built on GLTR rank features found that a word outside the top-100 predictions is over five times more likely in real text (odds ratio 5.32), whereas being the top prediction is far rarer (odds ratio 0.09) (arXiv:1906.04043). The authors further report that human-written text tends to use high-rank, non-obvious words regardless of predicted entropy, unlike generated text (arXiv:1906.04043).

The human-subjects study reinforces this: without the interface participants detected fake text at 54.2% accuracy, barely above chance, rising to 72.3% with GLTR (arXiv:1906.04043). Public uptake was substantial—30,000 demo page views and 21,000 blog views within the first month (arXiv:1906.04043). The open gap this leaves is robustness: the work validates a single statistical detector against its contemporaneous generators but does not establish how such rank cues hold under adversarial evasion or future model distributions.

中文

GLTR 最初是 2019 年的预印本 arXiv:1906.04043, 作者为 Sebastian Gehrmann (Harvard SEAS)、Hendrik Strobelt (IBM Research / MIT-IBM Watson AI Lab) 与 Alexander M. Rush (Harvard SEAS) (arXiv:1906.04043)。其核心论点是: 机器生成文本可通过统计线索被可靠检测并可视化供人检视——由于生成式语言模型从自身预测分布中采样词元, 生成文本将概率质量集中在高排名词元上, 统计检测器加可视化着色即可将其暴露 (arXiv:1906.04043)。

在实证层面, 基于 GLTR 排名特征构建的逻辑回归判别器发现, 位列前 100 之外的词出现在真实文本中的可能性高出五倍以上 (优势比 5.32), 而作为最高预测的概率则低得多 (优势比 0.09) (arXiv:1906.04043)。作者进一步指出, 人类书写文本无论预测熵高低都更常使用高排名、非显而易见的词, 这与生成文本不同 (arXiv:1906.04043)。

人类被试研究印证了这一点: 不使用界面时, 被试检测虚假文本的准确率仅为 54.2%, barely 高于随机; 使用 GLTR 后升至 72.3% (arXiv:1906.04043)。公众反响可观——上线首月内演示页获 30,000 次浏览、博客获 21,000 次浏览 (arXiv:1906.04043)。由此留下的开放缺口是鲁棒性: 该工作仅针对同期生成器验证了单一统计检测器, 未确立此类排名线索在对抗性规避或未来模型分布下的有效性。

Core Statistical Hypothesis

GLTR' s central hypothesis is that a language model leaves a statistical fingerprint in its own output: because generated text is sampled from the model' s predicted distribution, its tokens concentrate near the top of that distribution' s rank ordering. By scoring each token against a maintained language model and ranking it in probability space, the detector converts the generation process itself into evidence of provenance (arXiv:1906.04043). This rank signature is asymmetric to human writing: human text tends to use high-rank, non-obvious words regardless of the estimated entropy, whereas generated text clusters on the model' s most probable tokens (arXiv:1906.04043).

The hypothesis is validated both statistically and behaviorally. A logistic-regression discriminator built on GLTR rank features found that a word falling outside the top-100 predictions is over five times more likely to appear in real text (odds ratio 5.32), while being the top-1 prediction is comparatively rare (odds ratio 0.09) (arXiv:1906.04043). In a human-subjects study, participants without the GLTR interface detected generated text with 54.2% accuracy—barely above random chance—which rose to 72.3% when using GLTR (arXiv:1906.04043).

As established here by a single pre-GPT-3 source, the open gap is whether the token-rank hypothesis survives as generators improve: stronger sampling and higher-quality outputs may compress rank signatures, and the detector' s validity depends on its maintained scoring model remaining aligned with the generating model, leaving the hypothesis' s shelf-life unresolved (arXiv:1906.04043).

核心统计假设

GLTR 的核心假设是: 语言模型会在自身输出中留下统计指纹。由于生成文本是从模型自身预测分布中采样得到的, 其 token 会集中出现在该分布排序的顶端; 通过对每个 token 用维护的语言模型打分、并在概率空间中排序, 检测器便把生成过程本身转化为来源证据 (arXiv:1906.04043)。这种排序特征与人类写作不对称: 无论估计熵如何, 人类文本都更倾向使用高排名的非显然词, 而生成文本则聚集在模型最可能的 token 上 (arXiv:1906.04043)。

该假设同时得到统计与行为层面的验证。基于 GLTR 排序特征构建的逻辑回归判别器发现: 落在前 100 名预测之外的词, 在真实文本中出现的可能性高出 5 倍以上 (比值比 5.32); 而作为头号预测出现则相对罕见 (比值比 0.09) (arXiv:1906.04043)。在人类受试者研究中, 不使用 GLTR 界面时, 参与者识别生成文本的准确率为 54.2%——仅略高于随机猜测; 改用 GLTR 后升至 72.3% (arXiv:1906.04043)。

鉴于此处结论仅来自单一的前 GPT-3 来源, 遗留的开放问题是: 随着生成模型能力增强, token 排序假设能否继续成立——更强的采样与更高质量的输出可能压缩排序信号, 而检测器的有效性又依赖其维护的打分模型与生成模型保持对齐, 因此该假设的时效性尚属未决 (arXiv:1906.04043)。

The Detector Design

GLTR’s detector rests on a statistical premise: because generative language models sample tokens from their own predicted distributions, generated text concentrates probability mass on high-rank tokens, whereas human-written text tends to use high-rank (non-obvious) words more frequently regardless of the estimated entropy (arXiv:1906.04043). Mechanically, the detector runs every token through a language model and records its rank among the predicted vocabulary probabilities, producing a per-token signal of how “model-typical” a sequence is; the discriminative value of that design is precisely that human text prefers high-rank words even at low predicted entropy (arXiv:1906.04043).

These per-token ranks are then aggregated into classification features. A logistic-regression discriminator built on GLTR rank features found that a word outside the top-100 predictions is over five times more likely in real text (odds ratio 5.32), while being the top-1 prediction is far rarer (odds ratio 0.09) (arXiv:1906.04043). The resulting baseline detector outperforms untrained human judgment: participants without GLTR identified fake text at 54.2% accuracy, barely above chance, and improved to 72.3% when using the interface (arXiv:1906.04043).

The reported statistics, however, quantify the rank signal for a specific detector-generator pairing. How the per-token rank signal behaves when the detector model diverges from the generating model family — the design’s generalization boundary — is left unquantified by these results; that is the gap the current evidence leaves open.

中文速览：GLTR 检测器建立在逐 token 排名信号之上——生成式模型从自身预测分布采样，使生成文本的 token 集中在高排名词上，而人类文本无论预测熵高低都更倾向使用高排名（非显而易见）词（arXiv:1906.04043）。这些逐 token 排名被聚合为分类特征：逻辑回归判别器显示，位于前 100 预测之外的词在真实文本中的出现可能性高 5 倍以上（比值比 5.32），而身为第一预测的词则罕见得多（比值比 0.09）（arXiv:1906.04043）。该基线检测器优于未经训练的人类判断：不用 GLTR 界面时受试者识别假文本准确率为 54.2%，仅略高于随机，使用后升至 72.3%（arXiv:1906.04043）。然而报告结果仅量化了特定检测-生成模型组合下的信号强度；当检测模型偏离生成模型家族时该信号如何变化仍是未回答的空白。

Visualization for Human Inspection

GLTR (Gehrmann et al., 2019) makes machine-text detection human-inspectable by coloring each token according to its probability rank, so passages whose tokens sit almost entirely in the top positions stand out at a glance (arXiv:1906.04043). The underlying statistical rationale is that generative models sample from their own predicted distributions, concentrating probability mass on high-rank tokens; a logistic-regression discriminator built on GLTR’s rank features reports an odds ratio of 5.32 for words outside the top-100 predictions (reflecting real text) and only 0.09 for top-1 predictions, making them far rarer in real prose (arXiv:1906.04043). Human writing, in contrast, favors high-rank, non-obvious words regardless of the model’s predicted entropy (arXiv:1906.04043).

This histogram-based rendering converts statistical detection into a forensic display that non-experts can read without technical knowledge (arXiv:1906.04043). In a human-subjects study, accuracy detecting fake text rose from 54.2% without GLTR to 72.3% with it (arXiv:1906.04043); within its first month the demo logged 30,000 and the blog 21,000 page views (arXiv:1906.04043). Because this section rests on a single 2019 study, no cross-source conflict arises, but an open gap remains: whether token-level visualization scales to contemporary large language models and generalizes across languages is unverified here.

GLTR (Gehrmann et al., 2019) 通过按概率排名为每个 token 着色，使机器生成文本的检测可供人眼直接审阅：当一段文本的 token 几乎全部落在排名前列时，读者一眼即可察觉（arXiv:1906.04043）。其统计依据在于，生成式语言模型从自身预测分布中采样，将概率质量集中于高排名 token；基于 GLTR 排名特征训练的逻辑回归判别器显示，排名在 Top-100 之外的词对真实文本的几率比 (odds ratio) 为 5.32，而排名第一的词仅为 0.09，在人类文本中要罕见得多（arXiv:1906.04043）。相比之下，人类写作更倾向于使用高排名、非显而易见的词，与模型预测的熵

值无关 (arXiv:1906.04043)。

这种基于直方图的渲染将统计检测结果转化为可解释的取证视图，无需技术背景的最终使用者即可据此判断文本来源 (arXiv:1906.04043)。在受试者研究中，不使用 GLTR 时识别人工文本的准确率为 54.2%，使用后升至 72.3% (arXiv:1906.04043)；上线首月，演示页面与博客分别获得 30,000 与 21,000 次浏览 (arXiv:1906.04043)。由于本节仅依赖 2019 年这一项研究，不存在跨来源的冲突，但留下的空白依然明显：token 级可视化能否扩展到当代大语言模型、并跨语言泛化，在本来源中尚未得到验证。

Impact and Legacy

GLTR supplied early empirical evidence that statistical distributional features outperform human intuition at spotting neural text. Without the interface, participants detected fake text at 54.2% accuracy—barely above chance—which rose to 72.3% with GLTR (arXiv:1906.04043). A logistic-regression discriminator using GLTR rank features corroborated the signal: a word outside the top-100 model predictions is over five times more likely in real text (odds ratio 5.32), whereas a top-1 word is far rarer (odds ratio 0.09) (arXiv:1906.04043). The mechanism is that human writing favors high-rank, non-obvious words regardless of predicted entropy, unlike generated text (arXiv:1906.04043).

Combining this transparent statistical baseline with human-inspectable coloring proved a reference design, echoed throughout subsequent deepfake-text detection literature. The demo’s first-month traction—30,000 page views for the demo and 21,000 for the blog—signaled real demand (arXiv:1906.04043).

The open gap: the evidence rests on a single 2019 study of GPT-2-era models. Whether the 54.2%→72.3% human gain and the rank-based statistics persist against modern LLMs, and whether later detectors actually inherited this design, remains unverified; closing it requires synthesis of the downstream AI-text-detection literature.

中文

GLTR 提供了早期实证：统计分布特征在识别神经文本上胜过人类直觉。无界面时，受试者识别假文本的准确率仅 54.2%，略高于随机水平；使用 GLTR 后升至 72.3% (arXiv:1906.04043)。基于 GLTR 排序特征的逻辑回归判别器印证了该信号：居于模型前 100 名预测之外的词在真实文本中出现概率高出 5 倍以上 (比值比 5.32)，而位居首位预测的词则罕见得多 (比值比 0.09) (arXiv:1906.04043)。其机制在于人类写作无论预测熵高低都更偏好高排序、非显性词，与生成文本不同 (arXiv:1906.04043)。

透明的统计基线配合可人工检视的着色设计成为后续深度伪造文本检测器竞相参照的模板。演示上线首月即获得 3 万次页面浏览、博客 2.1 万次浏览，印证了实际需求 (arXiv:1906.04043)。

遗留空白：以上证据仅来自 2019 年基于 GPT-2 时代模型的单一研究。准确率增益与排序统计在当代大模型下是否依然成立、后续检测器是否真正沿袭此设计，均未验证，需进一步综合下游文献方能作答。

Conclusion / 结语

GLTR’s core result is that machine-generated text is *not* statistically invisible: because a generative LM samples tokens from its own predicted distribution, generated text concentrates probability mass on high-rank tokens, a fingerprint that both machines and humans can exploit. The quantitative evidence is sharp. A logistic-regression discriminator over GLTR’s rank features finds a word outside the top-100 predictions is over $5\times$ more likely in real text (odds ratio 5.32, vs. 0.09 for top-1 predictions), and human writing leans on high-rank, non-obvious vocabulary regardless of the model’s estimated entropy (arXiv:1906.04043). The signal survives model access: the same rank features separate real from generated text even without the true generating model (arXiv:1906.04043), and it transfers to people—naive participants detected fake text at only 54.2%, barely above chance, but reached 72.3% once GLTR’s ranked-annotation interface made the cues legible, with no prior training (arXiv:1906.04043). The post-publication traction (30,000

demo / 21,000 blog views within the first month) marks this legibility as a design element in its own right, not just an accuracy trick (arXiv:1906.04043).

The takeaway is twofold: (1) detection does not require a black-box classifier —a transparent, top- k / rank-based statistical visualization can convert an implicit distributional signal into an explicit human judgment; (2) the approach is a *moving target*, validated on GPT-2/BERT-era generators, English-centric data, and a modest human-subject study.

Remaining gaps / 未覆盖的缺口

- **Adversarial robustness (对抗鲁棒性)**: no evidence of calibration moderating causal acuity — GLTR is not stress-tested against *rewritten* or *prompt-steered* generation designed to escape top- k statistics; under adaptive attackers the heuristic signal may erode (*unverified*, out of scope for arXiv:1906.04043).
- **Generator generality (生成器泛化)**: the odds-ratio and entropy findings are tied to GPT-2/BERT-era LMs; modern decoding strategies, fine-tuned “human-calibrated” outputs, and multilingual data are untested.
- **Human-study scale (人类评估规模)**: the 54.2% $\rightarrow$ 72.3% gain rests on a single small-sample task design; no error ceiling, cross-domain replication, or causal account of *why* the interface helps has been established.
- **Downstream guarantees (无下游保证)**: GLTR yields a statistical fingerprint, not a proof — evidential-admissibility, long-term model drift, and detection-utility decay all remain open.

中文速览: GLTR 的核心启示是, 机器生成文本在语言模型的分布下并非“无可检测”——生成文本会把概率质量集中在前列 token, 该统计痕迹既可为机器所用, 也可被人类读取。实证上, 基于排名特征的逻辑回归判别器发现: 真实文本中出现“模型 top-100 预测之外”词汇的可能性是生成文本的 5 倍以上 (比值比 5.32, 而 top-1 预测词的比值比仅 0.09), 且人类写作无论熵估计高低都更偏好高排名词 (arXiv:1906.04043)。该信号在缺少真实生成模型访问权限时依然有效; 将其做成带排名标注的可视化界面后, 未经训练的受试者检测准确率由 54.2% (接近随机) 提升至 72.3% (arXiv:1906.04043)。发布首月 demo 与博客即获 3 万与 2.1 万独立浏览量, 说明“可读性”本身即是该方法的价值 (arXiv:1906.04043)。主要缺口: (1) 未对抗改写/提示对抗等自适应攻击做鲁棒性测试; (2) 结论局限于 GPT-2/BERT 时代生成器, 未覆盖现代解码策略、微调“拟人化”输出与多语言场景; (3) 人类实验规模小、缺乏跨域复现与因果解释; (4) 该方法是统计指纹而非证明, 证据采信与长期漂移问题未论及。

Evidence Table

#	Claim (abridged)	Source	Confidence
1	In a human-subjects study, participants without the GLTR interface detected fake text with 54.2% accuracy, which rose to...	arXiv:1906.04043	high
2	A logistic-regression discriminator using GLTR rank features found a word outside the top-100 predictions is over 5 time...	arXiv:1906.04043	high
3	Human-written text tends to use high-rank (non-obvious) words regardless of predicted entropy, unlike generated text.	arXiv:1906.04043	high
4	Within its first month of public deployment, GLTR’ s demo received 30,000 page views and its blog 21,000.	arXiv:1906.04043	high

References

- [\[1\]](#) arXiv:1906.04043